Job Recommendation and the Role of Stereotypes

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Abstract

Despite the active use of Job recommender systems, the reinforcement of social stereotypes remains an under-investigated issue. Prior work shows that gendered wording in job ads and biased Natural Language Processing (NLP) based algorithms can influence which candidates or jobs are recommended. The model introduces bias through both explicit values and those derived from the data, without any explicit specifications. In this research, we constructed an NLP-based job recommender system utilizing various datasets to investigate how different attributes of applicants, including specific writing stylistic parameters, influence the jobs for which they are recommended. Our goal is to identify where bias enters the recommendation process and suggest ways to mitigate it, which would also help us identify core features of each applicant responsible for their recommendation outcome. The results show that a significant amount of bias is introduced by simply adding noise as a biased sentence to the original prompt about the user. This signifies the potential liability of Sentence-BERT (SBERT)-style sentence embedding models, specifically the *JinaAI* embedding model used in this study, when exposed to biased or demographically salient input data.

1 Introduction and Motivation

In recent years, job recommender systems have become a prevalent algorithm that users interact with while searching for vacancies or candidates. While this simplifies the item-retrieval process by showing the most relevant items to both parties, those systems are prone to both algorithmic bias and stereotype leakage during training on potentially biased data [5]. In this research, we investigate the degree to which job recommender systems are biased and the underlying causes of this bias. We examine this problem by constructing an NLP-based job recommender system and by analyzing the process and results we encounter. This will show to what extent current pre-trained models, such as *SBERT*-style sentence embedding model [19], are biased.

Therefore, after meticulous analysis of existing research papers, we propose the following questions for further investigation, which are underexplored in the literature:

(1) **RQ1.** How does an alteration of the data input affect bias?
(2) **RQ2.** How biased is the model with respect to different introduced bias attributes such as gender?
(3) **RQ3.** How is bias distributed with respect to abstract text-features, such as readability and personality?

We will implement job recommenders based on publicly available datasets. It is discussed that the dataset affects the outcome of a corresponding Job Recommender System (JRS) [20]; therefore, we use two models with slightly different architectures to examine how data affects bias. Proposed datasets include the RecSys 2016 and RecSys 2017 competitions ([2] and [3], respectively), as well as the CareerBuilder 2012 Job Recommendation Challenge [10], which was hosted by CareerBuilder on Kaggle. We will work with the latter as it contains the most suitable data for our research.

The vast majority of publicly available datasets do not contain information that could be biased [1, 4, 10, 12]. Therefore, the vast majority of researchers in this field are focused on optimizing distance calculation [15] and time relevance [13]. With limited information provided, it does not allow for an exploration of bias, as such attributes are lacking in publicly available datasets. As this does not contain much bias, it renders the task closer to a black box. The closest research to this we found is a research by Zhang and Kuhn [20], which investigates JRS exactly from a black box standpoint. CF does not include data points, and it is complicated to generate artificial data that could introduce bias, since the training data had already been gathered. Therefore, we introduce two methods to explore LLM bias reinforcement.

(1) User-feature-based Content-Based Filtering (*CBF-U*) that takes into account the user's limited personal features, and
(2) Item-based Content-Based Filtering (*CBF-I*) that treats a user based on a list of items he has previously interacted with (in this case, those to which he applied).

We believe that investigating two models with different inputs creates a clearer overview of the bias. In particular, for *CBF-U*, which operates as an unsupervised model after removing its supervised component and reducing it to a pure similarity-based retrieval approach, we use another dataset containing job descriptions and salaries from LinkedIn Job Postings [12]. This allows us to examine bias further and explore how other features, such as salary, are affected by changes in gender in the user-vector. For the initial supervised model, we have a dataset with a limited number of attributes [10]. To explore bias, we introduce a random attribute called gender. We look at what jobs are recommended to users and what happens if we flip this attribute.

2 Related Work

Our investigation is mostly grounded in recent research about recommender systems based on NLP. Zhang and Kuhn [20] study gender bias in black-box job recommender systems. They interact with real job-board interfaces and compare recommendations for matched profiles that differ only in gender, which measures the effect of gender without needing access to the underlying models. At the same time, it doesn't investigate why the system behaves as it does, because internal features and ranking logic are hidden. The paper groups recommendations into 3 classes: **(Content-based) C-mechanisms** rely on profile content and rule- or NLP-based

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matching. **(Worker's action) W-mechanisms** rely on worker actions, such as viewing and applying to jobs. **(Recruiter's action) R-mechanisms** rely on recruiter actions, such as viewing or downloading a resume. Their results suggest that **C-mechanisms** can introduce gender gaps even before any user actions are taken, which, in our case, is interesting to investigate using embedding models. Prates et. al [17] suggests that this is because NLP similarity models can learn stereotypes from text and transfer it to matching and ranking. In addition, some systems may use declared gender directly as input feature, which can create differences even when qualifications are same. For **R-mechanisms**, they find that recruiter behaviour can amplify gaps when recommended jobs contain more stereotypically gender-coded content - a strong culture bias which changes perception based on masculinity/femininity of certain role. This, interestingly, affects the applicant side as well, as confirmed by [11], where experiments show why qualified women may avoid applying to higher-level roles. They show that small wording choices can activate meta-stereotypes, meaning beliefs about how others view women in the workplace. Women report lower job attraction when job descriptions include negative stereotypes (e.g. implying women are overly emotional). On the other hand, when ads include more positive, communal wording (for example, emphasizing cooperation and coaching), attraction and intention to apply increase. This matters for recommender systems because training signals come from observed behavior. If a candidate skips a recommendation due to biased choice of words, the system can interpret that as a lack of interest, even though the decision was driven by language rather than fit or similarity. This perspective is crucial for our study because it suggests that behavioral labels (applied / not applied) can impact the real accuracy of our recommender system - true user's preference may be impacted. Finally, [20] suggests **W-mechanisms**, that suggest evolution of bias through feedback loops. Implicit metrics such as clicks and applications often reflect the initial skew in recommendations, which essentially impacts further reinforcement of certain bias in the system. Overall, the study shows that measurable gender differences exist and that multiple mechanisms can contribute. Because it observes only inputs and outputs, it cannot test how changes in training data, prompt construction, or model components affect bias. In our work, we complement this research by building an embedding-based recommender where we change inputs in a controlled way and observe how those changes impact the output of our system.

Since our recommenders rely on sentence embeddings, we also build on work showing that NLP models can encode social stereotypes. Caliskan et al. [6] demonstrate that representations learned from text capture human-like associations, and they propose tests that quantify these effects. Later work shows that contextual embeddings also carry gender information present in the training data. [21] which directly confirms that untreated embedding models contain bias.

3 Methodology and Materials

In this section, we provide a detailed discussion on the methodology to ensure clear instructions for reproducibility. In this research, we have investigated two models, *CBF-U* and *CBF-I*. We generalize the overall methodology, making distinctions where necessary. If no mention is made of the difference, then two models are the same in this regard.

3.1 Datasets and Data Collection

In this work, we will only discuss the following data collections, which are presented in Table 1.

(1) **CareerBuilder** [10]. It includes users, job postings, and observed applications, organized into seven time windows with defined training and test periods. This structure provides ground-truth behavioral labels enabling the standard offline evaluation of ranking performance. The user table contains demographic and professional attributes. Possibly biased attributes, such as gender, age, and political beliefs, are lacking. While this is a positive aspect of the data set, we do not have true values for the bias investigation.
(2) **LinkedIn Job Postings (2023 - 2024)** [12]. It is a dataset with high-quality semantic descriptions of jobs containing textual fields (job title, description, requirements, and related metadata). This dataset is suitable for embedding-based retrieval as it provides natural-language content that can be transformed into dense embeddings for improved semantic matching. It supports analysis of language-driven bias, where gender-coded or age-coded cues in descriptions might influence which jobs appear most similar to a given user profile. Due to the high-quality semantic textual description of the jobs, it enables a deeper analysis of metastereotype-sensitive content, providing more insight into pre-trained models' analysis of the text.
(3) **RecSys Challenge 2017** [4]. It provides large-scale user–item interactions and structured attributes; however, it is anonymized and uses numeric concept IDs for most textual fields, which does not allow for an attribute analysis. While in practice this is a well-constructed dataset, these constraints render it useless for our controlled tests of gender and age-related recommendation differences made by textual profile or job content. This dataset is not used in this research.

We use the *CareerBuilder* dataset [10] for an initial training of the two models. We employ a temporal split during training, where the first 9 days serve as the training period, and the last 4 days are reserved for the validation set. For the test part, we use the LinkedIn job postings dataset [12], as it consists of data that can be explicitly used to interpret the results without human-perception subjectivity. The *CareerBuilder* consists of descriptions; there are no explicit attributes to write connections to the bias introduced. In contrast, the LinkedIn job postings dataset has attributes such as salary and experience level, which are used for a further correlation analysis with the introduction of an artificial bias.

3.2 Data Preprocessing

For the *CareerBuilder* dataset, we clean and standardize user and job metadata: remove duplicates, handle missing values by removing specific sentences during a prompt construction. For user history and job titles, we apply the text normalization, to be more precise: lowercasing, removing noticeable artifacts to build profile-text fields. The job descriptions in this dataset contain a high amount

Dataset	User Description	Job Description	Interactions
CareerBuilder	Attribute-based	Natural language-based	Yes
LinkedIn Job Postings	N/A	Natural language-based	N/A
ACM RecSys 2017	Attribute-based	Attribute-based	Yes

Table 1: Summary of datasets used in this study

of non-informative tokens, such as Hypertext Markup Language (HTML) code, which also need to be removed.

For the *LinkedIn Job Postings* dataset, we focus on text quality: we remove HTML/boilerplate, remove duplicates, and normalize whitespace. We then treat the generated text with the *SBERT*-style sentence embedding model *JinaAI* embedding model. For the synthetic personas, we generate templated profiles and then manually validate them with simple rule-based checks to ensure that paired profiles are identical in qualifications and interests, differing only in the targeted sensitive attribute. Furthermore, we balance the number of personas across attribute values to avoid side effects from unequal sample sizes. We exclude all items with *NaN* values in the description, title, normalized salary, company name, and formatted experience level to ensure that only records with complete information are used for comparison. As the description length varies substantially across the dataset, we further exclude all items with fewer than 250 words or more than 750 words in the description. To provide data about the description statistics: the median length is 280 words (rounded to the nearest ten), the upper quartile is 470 words, and the 95th percentile is 910 words.

Although job requirements are available (the attribute is named skill desc, but we refer to it as requirements for generalization), only approximately 300 items in the *CareerBuilder* dataset contain this information, which would severely limit the scope of our experiments. After a manual inspection, we observed that jobs without explicitly stated requirements typically include this information within the general description. Consequently, we exclude items with a separate requirements field and retain only those jobs where the requirements are embedded in the description.

Finally, we restrict the dataset to full-time positions with a yearly pay period and salaries denominated in U.S. dollars to avoid additional sources of variability. After applying all filtering criteria, the resulting dataset contains approximately 7,500 items used for evaluation.

3.3 Models

Due to the bias-investigation task, our models are limited to content-based filtering recommendation systems. The model's different structure does not incorporate the artificial attributes we add to its pipeline. While this negates explicit bias reinforcement, it still partially reinforces implicit bias, as suggested in a research paper by Zhang and Kuhn [20]. We consider the following four models in this research:

(1) **CBF-U**. Content-based Filtering that takes into account the user's limited personal characteristics. (CBF-U for short). The CBF-U pipeline is illustrated in Figure 1. The user prompt is constructed and then fed into an *JinaAI* embedding model, which is an SBERT-based model [19]. The low-dimensional representation of a user is put through n Feedforward Neural Network (FNN) layers. Finally, using cosine similarity, we determine which job vacancy best suits the user. The job is processed through the same pipeline of sentence construction, utilizing existing features such as a description, title, salary, and required experience level. Then all the sentences created are fed to an LLM (JinaAI embedding model) to produce a low-dimensional vector.

(2) **CBF-I**. Content-Based Filtering that treats a user based on a list of items he has previously interacted with (in this case, those to which he applied). (CBF-I for short). CBF-I is visualized in Figure 2. We generate k amount of sentences using natural-language features about a job. The jobs for which the user applied are split in half, with one half used for the user representation and the other used for training to maximize cosine similarity between the user representation and target jobs. All the generated sentences are put into the *JinaAI* embedding model. An average of all representations is a representation of a user. Finally, as in the previous baseline, we use cosine similarity to determine which job vacancy best suits the user. The job vacancies data is initially preprocessed as described in the CBF-U baseline. Given that the user is being embedded based on their application history, we explore the nearest matches to the embeddings generated for each job from the LinkedIn dataset.

The style of a prompt fed to *JinaAI* embedding model heavily relies on the initial prompt provided by a dataset and on the number of artificial attributes. The prompt looks as follows, for the training data-users for the *CareerBuilder* dataset:

> I have a {DegreeType} education, where I got a major in Major in {GraduationYear}. I have {TotalYearsExperience} years of experience and have held {WorkHistoryCount} roles. Work history includes: {JobTitle1}; {JobTitle2}; … . I am (not) currently employed. I have people-management experience. I have people-management experience leading a team of {ManagedHowMany} people. My gender is Male/Female. (A randomly generated token added during training, for this research, we state it as a binary variable.)

It is essential to note that all fields in the prompt are conditionally included based on the availability of data. Because all parts of the user's data are optional, we encounter a large number of users who may miss certain parts of their profiles. Education information is omitted when both "degree type" and "major" are missing. Work history details are omitted only if no prior role of a person is present. The job titles are listed in chronological order with duplicates removed, which is a common case. Employment information is omitted if not available. Management information is

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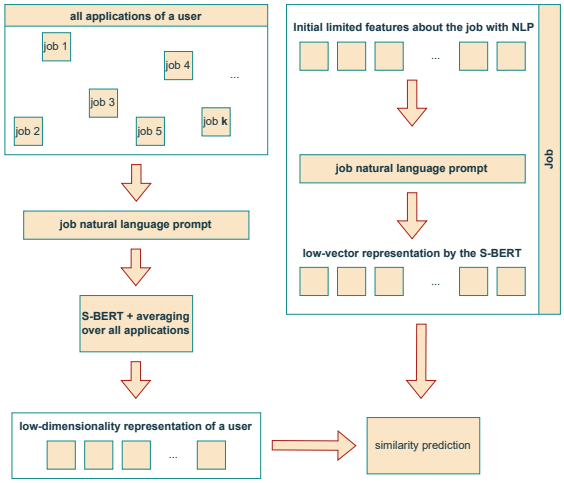

Figure 1: Pipeline overview of a CBF-I model

expressed either as a generic people-management experience or as team-size–specific leadership experience. We construct data this way to avoid introducing artificial or low-information attributes.

For a task simplification, we do not include explicit distance calculation in our baselines. After a manual inspection of the data, we observed that most users apply for jobs across the country (mostly within the United States), rather than being restricted to a single state. Therefore, while location information is informative, it is not crucial, and we omit potential location bias. The training data-items for the *CareerBuilder* dataset, the prompt looks as follows:

> Title: {Title}. Description: {Description}. Requirements: {Requirements}.

The median text length is approximately 280 words (rounded to tens), the upper quartile is 470 words, and the 95th percentile is 910 words. To retain most of the content while still utilizing the *SBERT*-style sentence embedding model structure, which is common and practical for the given task, we use the *JinaAI* embedding model, as implemented in the Hugging Face Transformers library [1]. The *JinaAI* model supports sequences of up to 8192 tokens. It is based on a *BERT* [8] architecture (*JinaBERT*) that supports the symmetric bidirectional variant of *ALiBi* [18] to allow longer sequence length. We limit the sequence length to 2048 for the job embeddings and to 1024 for the user embeddings. We base this restriction on the average number of words in that dataset.

The testing data items from the *LinkedIn Job Postings* dataset, the prompt looks as follows:

> Title: {Title}. Description: {Description}. Requirements: {Requirements}. Salary per year: {Normalized yearly salary} dollars. Required experience level: {Experience level required}.

[1]https://huggingface.co/docs/transformers/index

4 Experimental Setup - Implementation

We first preprocess the CareerBuilder interaction dataset and construct two user representations: (1) attribute-based profiles (CBF-U) built from available user features and (2) history-based profiles (CBF-I) built from previously applied jobs. For each user and job, we generate short natural-language summaries and create parallel text representations using *JinaAI* embedding model.

4.1 Data

As stated before, our work primarily consists of experiments with textual representations using the *JinaAI* embedding model, specifically *jina-embeddings-v2-small-en*, which yields an embedding dimension of 512. For both pipelines, representations were pre-calculated on cleaned data. Specifically, user-prompts and job-prompts from the CareerBuilder dataset, as well as job prompts from the LinkedIn dataset.

4.2 CBF-I

The interaction model works by manipulating user interaction history to build a single profile, without any specific user attributes.

$$\mathbf{h}_u = \frac{1}{k}\sum_{i=1}^{k}\mathbf{v}_{j_i}$$

where $\mathbf{h}_u$ is the user representation computed as the average of k job vectors. Here, $\mathbf{v}_{j_i}$ is the embedding of the i-th job (with index j_i) selected from the user's interaction history, and k is the number of those jobs included in the average.

For the baseline, we extract top-20 matching job embeddings using similarity from the LinkedIn dataset:

$$\mathrm{sim}(u, j) = \frac{\mathbf{e}_u \cdot \mathbf{v}_j}{\|\mathbf{e}_u\|\|\mathbf{v}_j\|}$$

Both vectors are L2-normalized before computing the dot product, making cosine similarity equivalent to the inner product.

For bias investigation, we create paired user embeddings that are identical except for an injected gender signal controlled by a parameter α. For each user, we retrieve the top-20 recommendations under both male and female conditions and compare the resulting job attributes—primarily salary, but also job title distributions and industry sectors.

$$\mathbf{e}_u^{(\mathrm{gender})} = \mathrm{normalize}(\mathbf{h}_u + \alpha \cdot \mathbf{g})$$

where g is the gender direction vector and α controls injection strength.

4.3 CBF-U

We do not change our function in any way during training. We transform all our prompts into 512-dimensional embeddings using JinaAI. Then we use training data to enhance the model with supervised data, we train a *UserAdapter*, a simple 2-layer FNN with 512 as the hidden layer dimensionality. Then we use validation data for early stopping. Next, we create synthetic persona pairs that are identical in qualifications and interests but differ only in sensitive attributes, under both attribute-sensitive and attribute-neutral prompting. For that, we pick users' data that was not used for the training randomly and add biased sentences. We run the

same retrieval/ranking pipelines and visualize how flipping sensitive attributes changes recommended salary-related outcomes, and compare stability across embedding methods and prompt variants.

4.4 Evaluation Protocol and Metrics

The initial task focuses on bias exploration; therefore, the main objective is to gather predictions correlated with respect to the biased artificial attributes. We will introduce n copies of the same profiles, differing only in an artificial attribute, and collect data for further visual investigation with respect to salary, pay period, position, and other attributes. Therefore, the goal of this paper is to provide a visualization for correlations between specific biased attributes, such as gender, with other non-artificial attributes.

4.4.1 CBF-I. Our evaluation focuses on three key metrics:

(1) Salary gap: The difference in mean recommended salary between male-injected and female-injected embeddings, tested for statistical significance using paired t-tests with 95% confidence intervals.
(2) Effect size: Cohen's d to quantify the magnitude of any observed salary differences, with standard thresholds (0.2 = small, 0.5 = medium, 0.8 = large).
(3) Recommendation overlap: The percentage of jobs appearing in both male and female top-20 lists, indicating how much gender injection changes the actual recommendations rather than just their salary characteristics.

We vary α from 0 (neutral baseline, no gender signal) to 2.0 (strong gender signal) to understand how increasing the strength of gender in embeddings affects recommendation outcomes. This approach allows us to quantify potential gender-salary correlations in the embedding space in an unsupervised setting.

4.4.2 CBF-U. We use cosine similarity as the loss function to maximize the similarity between users and the items they have interacted with, while minimizing this value with respect to other values. For the evaluation, as our task is not to improve the model indefinitely, but to research bias, we use correlations with other explicit attributes. Our primary focus is to identify the effect of a specific noise signal on the prediction; therefore, we construct two tables: Table 2, comprising various bias introductions and the corresponding effects of this data included in the candidate information. This table explores the correlation between the noise signal introduced (bias signal) and salary, and Table 3 shows the correlation with bias signal and readability and personality attributes, derived from the text.

4.5 Preprocessing and Train/Test Splitting

Our data splitting strategy will follow the structure of the Kaggle dataset (CareerBuilder) challenge as it supports temporal generalization by default. Each window already contains a 9-day training period and a 4-day test period, and users/jobs are assigned to exactly one window. We will train interaction-based models on the provided training portion of each window and evaluate on the official test portion for that window. To tune hyperparameters and compare methods fairly, we create a validation set by using one full window (e.g., Window 2) as the development set.

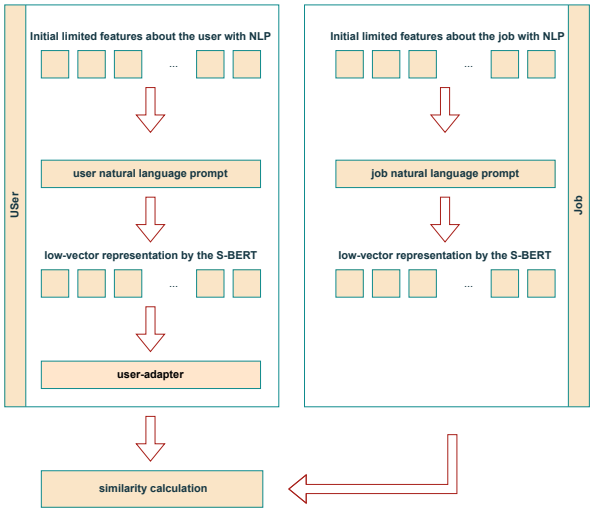

Figure 2: Pipeline overview of a CBF-U model

For the LinkedIn text corpus, since we will primarily use it for job-text embeddings rather than supervised interaction prediction, we do not need a classic train/test split; instead, we use it uniformly across experiments to compare how different persona variants retrieve different job sets. The synthetic personas are also split into development and evaluation lists. This design is aligned with our research questions. Temporal splits in the interaction dataset are important because job applications are time-sensitive and because we want to measure realistic ranking performance before analyzing fairness effects. We need to ensure that we do not create or confuse anything else before we analyze the biases.

5 Results

5.1 CBF-U

Table 2 presents a comparison of the average salary and the mode of experience level with respect to the randomly introduced biased sentence. The different groups exhibit a relatively large number of differences. We observe that simply adding a sentence specifying a particular attribute to the model, regardless of the attribute's value, already affects the results. For example, when a "Language proficiency" or "Political alignment" statement is present in the profile, irrespective of its value, the average recommended salary increases noticeably. Mode Experience is not affected by the bias.

Table 3 presents a comparison of the personality scores and a readability score (specifically, Flesch Reading Ease [9]), a metric estimating the structural complexity of a text and the level of literacy required to understand it. We model user personality using the five-factor (OCEAN) model, which comprises Extroversion, Neuroticism, Agreeableness, Conscientiousness, and Openness [14]. Personality traits are inferred from users' textual content using a publicly available BERT-based model[2] fine-tuned for personality

[2]https://huggingface.co/Minej/bert-base-personality

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Bias Category	Group	Avg. Salary	Mode Experience
Baseline (no demographic attribute)			
None	none	109,000	Mid-Senior level
Religious identification			
	non-religious	97,000	Mid-Senior level
	religious	98,000	Mid-Senior level
Age group			
	18–28	110,000	Mid-Senior level
	29–42	111,000	Mid-Senior level
	43–64	105,000	Mid-Senior level
Gender			
	female	102,000	Mid-Senior level
	male	110,000	Mid-Senior level
	non-binary	112,000	Mid-Senior level
Political alignment			
	conservative	126,000	Mid-Senior level
	liberal	141,000	Mid-Senior level
	moderate	131,000	Mid-Senior level
Language proficiency			
	native speaker	122,000	Mid-Senior level
	non-native speaker	120,000	Mid-Senior level
Disability status			
	has a disability	106,000	Mid-Senior level
	no disability	109,000	Mid-Senior level

Table 2: Bias audit results for the recommendation system. Average normalized salary (rounded to thousands) and most frequent experience level among top-K recommended jobs. The K is 50, the number of users used for every group is 5000, and the number of possible job recommendations is 7500. The baseline corresponds to synthetic profiles with no demographic attribute added.

Bias	Value	Read.	Extr.	Neur.	Agree.	Consc.	Open.
None	none	6.67	0.13	**0.04**	0.20	0.13	0.33
Religious identification	non-religious	3.94	0.11	0.03	**0.24**	**0.17**	0.32
	religious	6.71	0.10	0.03	0.23	0.16	0.32
Age group	18–28	3.30	0.12	**0.04**	0.21	0.15	0.33
	29–42	2.27	0.12	**0.04**	0.21	0.15	0.33
	43–64	4.56	0.11	0.03	0.22	**0.17**	0.32
Gender	female	**7.58**	0.12	0.02	0.22	**0.17**	0.32
	male	5.46	0.12	0.03	0.21	0.14	0.32
	non-binary	4.44	0.12	**0.04**	0.21	0.14	0.32
Political alignment	conservative	5.90	0.13	**0.04**	0.23	0.14	0.34
	liberal	5.97	**0.14**	**0.04**	0.22	0.14	**0.35**
	moderate	3.76	**0.14**	**0.04**	0.22	0.12	0.34
Language proficiency	native speaker	5.95	0.13	**0.04**	0.18	0.11	0.34
	non-native speaker	5.24	0.13	**0.04**	0.19	0.13	0.34
Disability status	has a disability	2.07	0.09	**0.04**	0.21	0.16	0.32
	no disability	0.73	0.09	**0.04**	0.21	0.16	0.32

Table 3: Average readability and personality scores of recommended job descriptions across bias conditions. Bold indicates the highest value per column; underlined values indicate the second-highest. The statistics are calculated across all jobs.

prediction, where BERT provides contextualized text representations [7]. From a logical standpoint, we would expect a certain correlation between salary and personality. The results show low differences overall. The largest differences appear in the readability score; however, comparisons across different bias categories are not informative, and within each bias condition, the observed correlation is relatively small.

5.2 CBF-I

Figure 3 displays the analysis of how the strength of bias injection affects the user representation, presented in two panels. Panel A displays absolute salary values, while Panel B shows the change in salary with respect to the strength of the injected signal. Positive values indicate that male-gender-injected jobs have higher salaries, while negative values indicate the opposite. The results indicate a noticeable degree of change, albeit in moderate amounts. Another observation is that the recommended item average salary increases with the introduction of a gender signal.

Figure 4 further examines the distribution difference in job salaries with respect to the strength of the injected signal. A clear observation is the increase in variance with the increased strength of the injected signal. This is expected, as with the signal, we introduce noise that does not correlate with the recommendations.

Figure 5 shows the recommendation overlap between male and female embeddings for Top-20 recommendations. At 0, as we have the same job, the overlap is 0. With the increase in noise, as we inject different signals into the same user representation, the overlap decreases, stagnating at 60%, indicating that the original user signal is preserved to some degree.

6 Discussion

RQ1. *How does an alteration of the data input affect bias?*

To investigate this question, we created two distinct models that take different inputs during training while addressing the same task. CBF-U represents a user through a prompt generated from personal attributes, including degree type, years of experience, management experience, graduation year, work history expressed as job titles, and current employment status. In this setting, bias is introduced explicitly as an additional sentence appended to the end of the prompt. While we acknowledge that such a formulation may make the model sensitive to the placement of biased information, limitations in computational resources prevented us from conducting more extensive prompt-structure ablations. The resulting bias effects are presented in section 5.

As an alternative, while still utilizing the same training data, the CBF-I model focuses on items and represents a user based on their interaction history. In this approach, bias is introduced by adding a gender embedding corresponding to a demographic attribute, scaled by a factor α, to the original user embedding, which is then scaled and normalized. The results show that on average, the interaction-based model (CBF-I) is more immune to bias, exhibiting a low difference in recommendations even with a high-valued α value.

RQ2. *How biased is the model with respect to different introduced bias attributes, such as gender?*

This question has been investigated using the CBF-I approach. We investigated the influence of the injected gender attribute by adding a scaled embedding to the user embedding, followed by normalization. Our results show that the interaction-based model is relatively robust to gender bias. When a user is represented by the mean of many job embeddings from their history, this aggregated signal is strong and stable. The added gender vector, even when scaled up, cannot easily override the established preference pattern. However, retrieval differences are still valuable.

RQ3. *How is bias distributed with respect to abstract text-features, such as readability and personality?*

For this model, we calculated Personality traits, a psychological model that uses a user's digital footprint to identify five broad personality dimensions: Extroversion, Neuroticism, Agreeableness, Conscientiousness, and Openness, commonly referred to as OCEAN [14]. Clear pattern recognition between input text and personality characteristics is achieved using a publicly available BERT-based model[3] that has been fine-tuned for personality prediction. Furthermore, we compute the Flesch Reading Ease, a metric that indicates the structural complexity of a text and the literacy level required to comprehend the information provided. The higher the value, the easier the text is to read. Our results indicate minimal variation in inferred personality traits across bias conditions, whereas differences in readability are more pronounced. The largest within-group difference is observed for the gender attribute: job descriptions recommended under the female condition exhibit a readability score of 7.58, compared to 4.44 for the non-binary condition. While some variation can be observed within individual bias categories, comparisons across different bias groups are not directly informative due to differences in the underlying semantics of the attributes.

7 Threats to validity – limitations

A primary challenge is the absence of certain user parameters that influence the study's outcome. The problem is that the *CareerBuilder* interaction dataset includes some demographic information. However, it does not fully cover the sensitive attributes that we want to test in a controlled way. To address this limitation, we augment user profiles by injecting randomly assigned sensitive attributes. This approach enables the construction of paired or grouped personas in which qualifications and professional interests remain fixed, while only the sensitive attributes are systematically varied. This design constrains the scope of real-world bias assessment. However, when such attributes are introduced in a controlled manner, it allows us to observe how SBERT-style sentence embedding models (specifically the JinaAI embedding model) alter their predictions in response to injected bias signals.

A further threat to validity arises from the scale of the dataset. Due to the significant computational cost associated with large-scale NLP processing, it was not feasible to utilize the entire dataset. The *CareerBuilder* data collection contains 1,091,923 job postings; however, only a subset was used in our experiments. This subsampling introduces potential variance in the results and limits the generalizability of the findings.

Additionally, the positional placement of sensitive attributes within the CBF-U model was not explored. As mentioned earlier,

[3] https://huggingface.co/Minej/bert-base-personality

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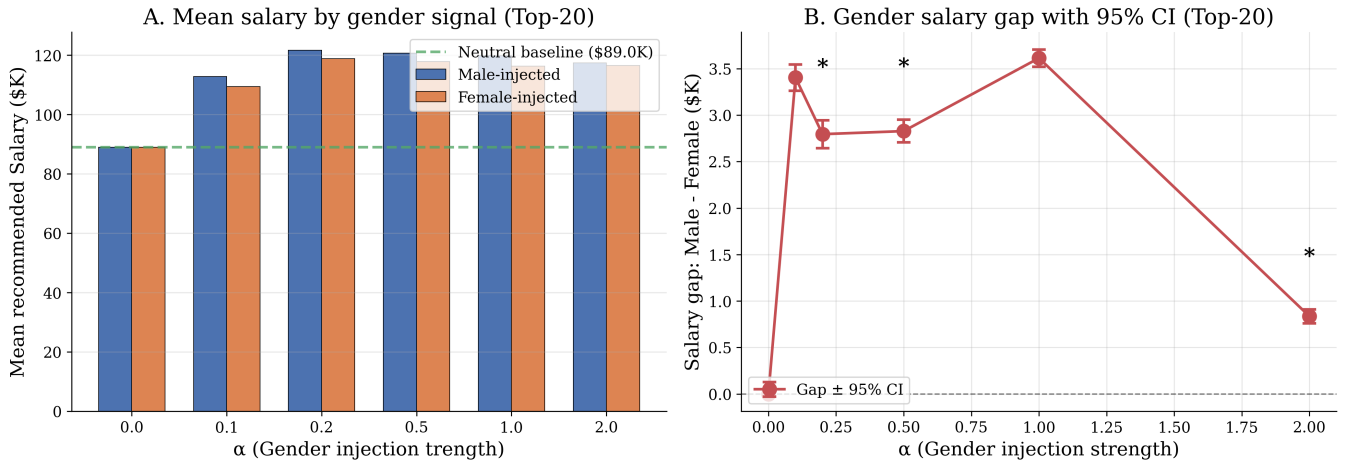

Figure 3: Gender bias analysis across different injection strengths (α). Panel A shows the mean recommended salaries for male and female employees. Panel B displays the salary gap (male minus female) with 95% confidence intervals. The neutral baseline ($\alpha = 0$) is highlighted in green. Significance is indicated with asterisks ($p < 0.05$, $^{*}p < 0.01$, $^{**}p < 0.001$).

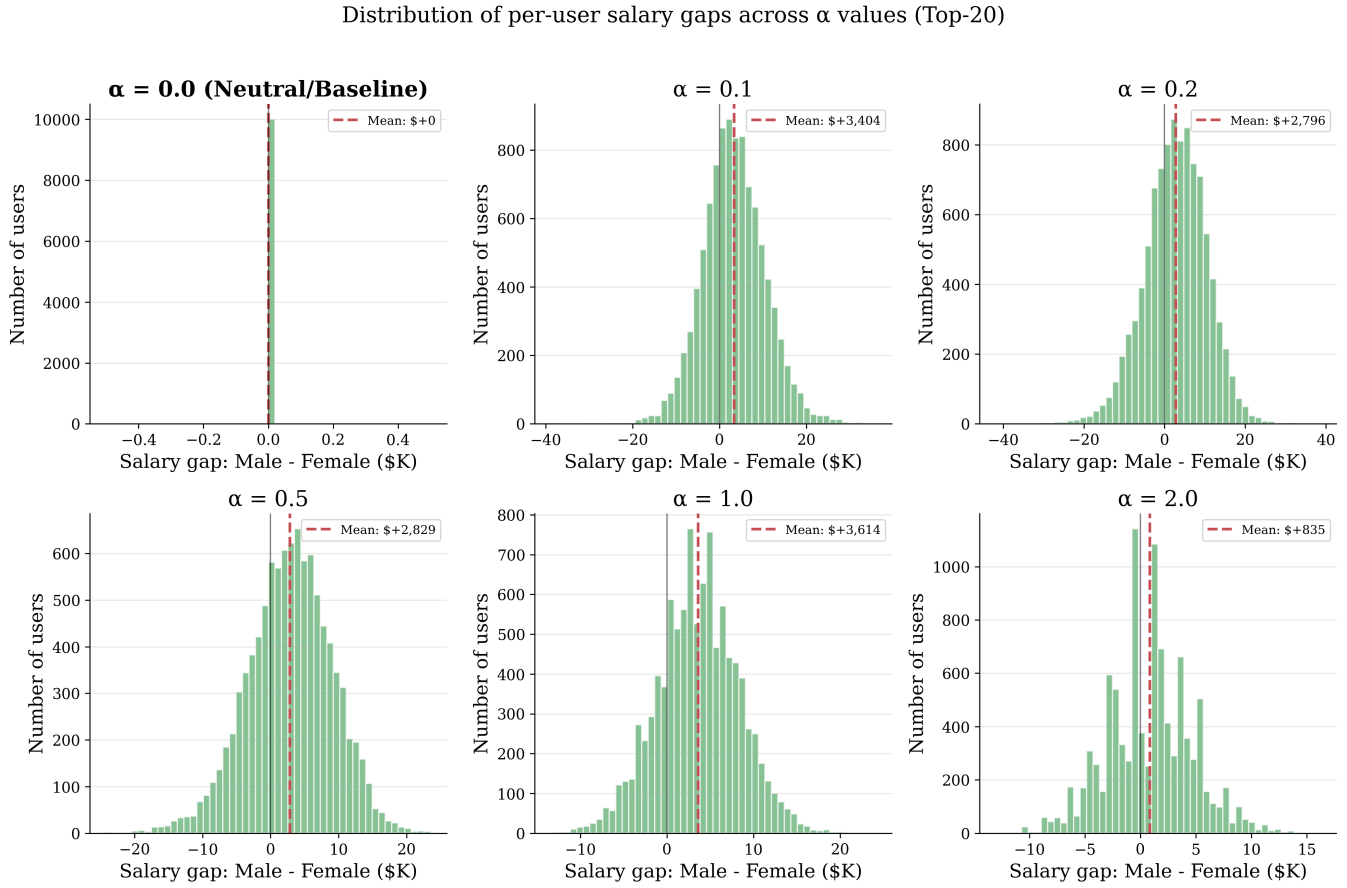

Figure 4: Distribution of per-user salary gaps across different α values. At $\alpha = 0$ (neutral baseline), the distribution is centered at zero since male and female embeddings are identical. As α increases, the distribution spreads, showing more variation.

Rank	Job Title	Salary	Level
Injected Male Bias			
1	Data/Configuration Management Specialist II	119,900	Entry level
2	Networks Information Assurance Test Engineer	119,900	Entry level
3	Functional Expert Consultant II	101,850	Mid-Senior level
4	AM Senior Manufacturing Engineer	**152,500**	Mid-Senior level
5	Solar Electrical Engineer – Hybrid	90,000	Mid-Senior level
Injected Female Bias			
1	Senior Manager Sourcing, Women's Lingerie	125,000	Mid-Senior level
2	Federal Regional Director of Sales (Remote, DMV)	**185,000**	Director
3	Senior Merchandise Manager Women	129,450	Mid-Senior level
4	Process Technician IV	68,750	Mid-Senior level
5	AM Senior Manufacturing Engineer	152,500	Mid-Senior level

Table 4: Top-5 job recommendations under gender-biased input injections. Bold indicates the highest value per injection; underlined values indicate the second-highest.

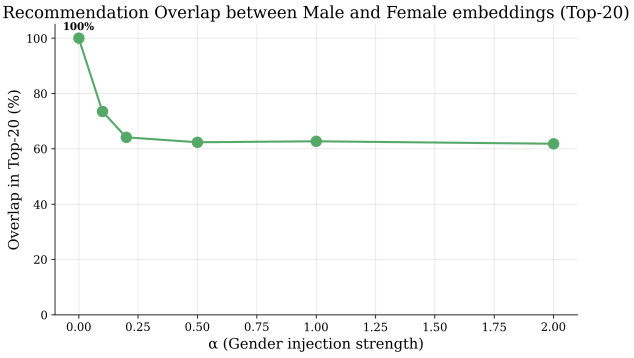

Figure 5: Recommendation overlap between male and female embeddings for Top-20 recommendations. At $\alpha = 0$, overlap is 100% (identical recommendations). Higher α values reduce overlap. Results represent averages across user profiles.

the placement may potentially alter the predictions negatively or positively. We leave this as future work to be explored.

8 Conclusion and future work

Our study has some limitations. The salary data on LinkedIn is incomplete, so not all jobs have salary information. This reduces the number of samples we can analyze. The gender signal we inject uses simple text prompts. Real-world systems may encode gender in more complex ways than we can fully capture. We also noticed that prompt text is very sensitive to word ordering. Small changes in how we write the user profile can affect the embeddings and, therefore, the recommendations. This means our results may depend on how we constructed the prompts.

Future research could explore other sensitive attributes like age or ethnicity. Developing methods to reduce bias while maintaining high-quality recommendations is also important. Another direction is studying how prompt construction affects results. Our embedding generation took around 10 hours for one time window of jobs, which limited our experiments. With more computational resources, systematically testing different prompt formulations would help understand how text representation influences bias in recommendations.

9 Disclosure of LLM assistance

We declare that a large language model GPT-5 [16] was used selectively to assist with programming-related tasks. No AI system was used to generate research ideas, define the experimental design, select models, draft the conclusions, or interpret results.

The Grammarly application was used for grammar checking and minor stylistic improvements. Tables were generated using the large language model GPT-5 [16] based on the computed experimental results.

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